

1	Course Name:
	Natural Language Processing
2	Course Code:
	IQB20404
3	Year and Semester Offered:
	Year 2 Semester 2
4	Credit Hours Value:
	4 Credit Hours [4 SLT Credit]
5	Prerequisite(s), if applicable:
	IQB10104 AI Programming ISB30503 Data Structures and Algorithms ISB47003 Machine Learning
6	Course Synopsis:
	This course provides an introduction to Natural Language Processing (NLP), focusing on computational methods for analyzing and generating human language. The course covers text processing, language modeling, syntactic and semantic analysis, and various approaches for NLP tasks. Students will develop applications for text classification, sentiment analysis, and conversational systems.
7	Course Learning Outcome:
	Upon completion of the course, students should be able to: 1. Explain the fundamental concepts, theories, and challenges in NLP. 2. Apply essential NLP techniques to process and analyze text data. 3. Build an NLP-based AI application and evaluate its performance using appropriate evaluation metrics.
8	Content Outline:
	Topic
	1. Introduction to NLP
	2. Text Processing: Tokenization, Normalization, Stemming, Lemmatization, Stopwords
	3. Part-of-Speech (POS) Tagging, Named Entity Recognition (NER)
	4. Syntactic Analysis: Parsing, Syntax Trees
	5. Semantic Analysis: Word sense Disambiguation, Topic Modeling, Sentiment analysis
	6. Language Models: n-grams, word representation, Word embeddings, Neural Language Models, Transformers, Document similarity measures
	7. Machine Learning for NLP: Supervised & Unsupervised Learning, Feature Engineering
	8. Deep Learning for NLP: RNNs, LSTMs, BERT
	9. Bias in AI and Explainability in NLP
9	References:

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| | <ol style="list-style-type: none">1. Jurafsky, D., & Martin, J. H. (2025). Speech and Language Processing (3rd Edition, draft). Pearson. [Online]2. Bird, S., Klein, E., & Loper, E. (2009). Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit. O'Reilly Media, [ISBN: 9780596516499]. |
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